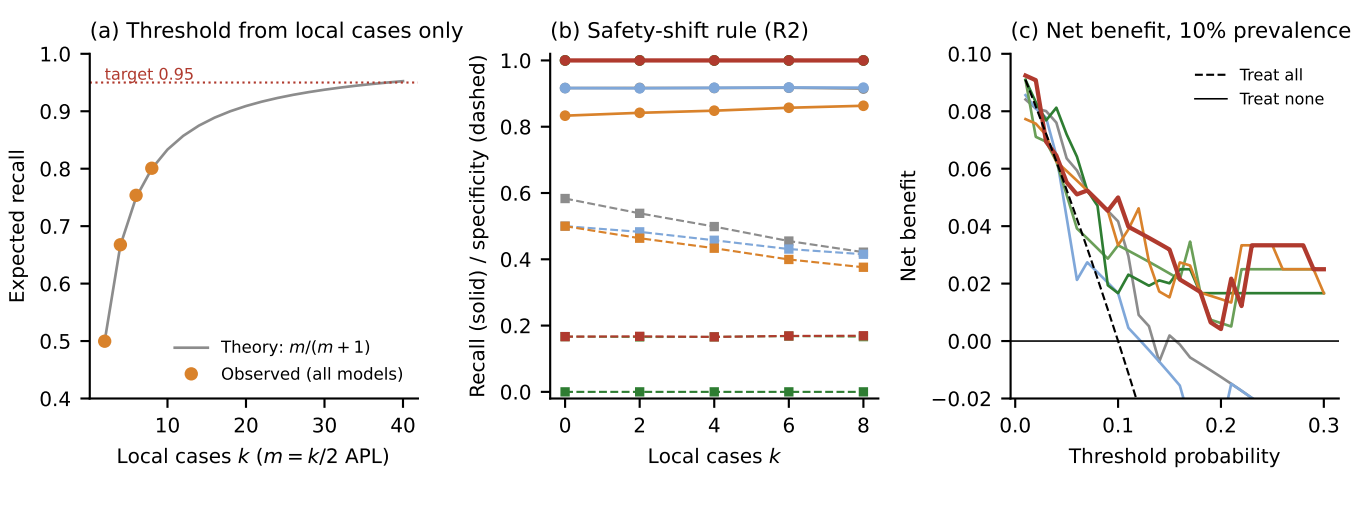
(a) Threshold from local cases only
target 0.95
Expected recall
Local cases k ($m = k/2$ APL)
Theory: $m/(m+1)$
Observed (all models)
(b) Safety-shift rule (R2)
Recall (solid) / specificity (dashed)
Local cases k
(c) Net benefit, 10% prevalence
Net benefit
Threshold probability
Treat all
Treat none
Macenko + frozen CNN, no PCA
Secondary: Macenko + frozen CNN + PCA-64
Engineered, pooled LR
Engineered, attention MIL
Macenko + fine-tuned EfficientNet-B0
Primary (ensemble)